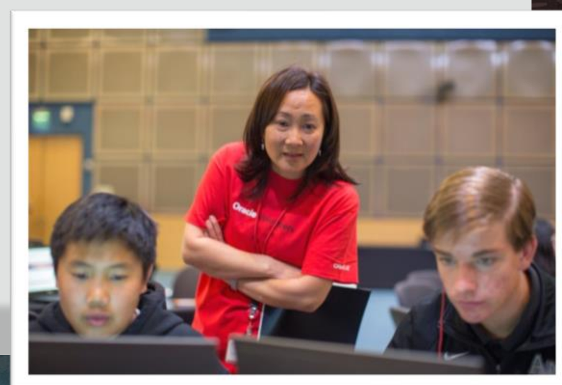
ORACLE
Academy

Database Programming with PL/SQL

7-1

Handling Exceptions

ORACLE
Academy



Objectives

- This lesson covers the following objectives:
  - Describe several advantages of including exception handling code in PL/SQL
  - Describe the purpose of an EXCEPTION section in a PL/SQL block
  - Create PL/SQL code to include an EXCEPTION section
  - List several guidelines for exception handling

Purpose

- You have learned to write PL/SQL blocks with a declarative section and an executable section
- All the SQL and PL/SQL code that must be executed is written in the executable block
- Thus far, you have assumed that the code works fine if you take care of compile time errors
- However, the code can cause some unanticipated errors at run time
- In this lesson, you learn how to deal with such errors in the PL/SQL block

Computer programs should be written so that even unanticipated errors are handled, no program should ever just crash or quit working.

What is an Exception?

- An exception occurs when an error is discovered during the execution of a program that disrupts the normal operation of the program
- There are many possible causes of exceptions: a user makes a spelling mistake while typing; a program does not work correctly; an advertised web page does not exist; and so on
- Can you think of errors that you have come across while using a web site or application?

What is an Exception?

- Some examples of errors you may have seen:
- Entering an incorrect username and/or password
- Forgetting to include the @ in an email address
- Entering a credit card number incorrectly
- Entering an expiration date that has passed
- Selecting more than one row into a single variable
- Receiving “no rows returned” from a select statement

Exceptions in PL/SQL

- This example works fine. But what if v_country_name was 'Korea, South' instead of 'Republic of Korea?'

```
DECLARE
  v_country_name countries.country_name%TYPE
   := 'Republic of Korea';
  v_elevation countries.highest_elevation%TYPE;
BEGIN
  SELECT highest_elevation
 INTO v_elevation
 FROM countries
 WHERE country_name = v_country_name;
  DBMS_OUTPUT.PUT_LINE(v_elevation);
END;
```

```
1950

Statement processed.
```

ORACLE Academy

PLSQL 7-1
Handling Exceptions

To confirm the spelling of Republic of Korea in the data, run a statement such as the following:

```
SELECT country_name
  FROM countries
  WHERE country_name LIKE '%Korea%';
```

Exceptions in PL/SQL

- When our v_country_name is not found, our code results in an error

```
DECLARE
  v_country_name countries.country_name%TYPE
    :='Korea, South';
  v_elevation countries.highest_elevation%TYPE;
BEGIN
  SELECT highest_elevation
 INTO v_elevation
 FROM countries
 WHERE country_name = v_country_name;
END;
```

```
ORA-01403: no data found
```

ORACLE Academy

PLSQL 7-1
Handling Exceptions

 8

Remember that a SELECT statement in PL/SQL must return exactly one row. This statement returns no rows and therefore raises an exception in the executable section.

Exceptions in PL/SQL

- The code does not work as expected
- No data was found for 'Korea, South' because the country name is actually stored as 'Republic of Korea'
- This type of error in PL/SQL is called an exception
- When code does not work as expected, PL/SQL raises an exception
- When an exception occurs, we say that an exception has been "raised"
- When an exception is raised, the rest of the execution section of the PL/SQL block is not executed

What Is an Exception Handler?

- An exception handler is code that defines the recovery actions to be performed when an exception is raised (that is, when an error occurs)
- When writing code, programmers need to anticipate the types of errors that can occur during the execution of that code
- They need to include exception handlers in their code to address these errors
- In a sense, exception handlers allow programmers to "bulletproof" their code

PL/SQL programs will start to get longer and more complicated due to the exception handling code needed to handle all errors, but that is preferable to programs just crashing or quitting. By including exception handling code, error messages can help the user understand what went wrong and how to correct it.

What Is an Exception Handler?

- What types of errors might programmers want to account for by using an exception handler?
- System errors (for example, a hard disk is full)
- Data errors (for example, trying to duplicate a primary key value)
- User action errors (for example, data entry error)
- Many other possibilities!

Why is Exception Handling Important?

- Some reasons include:
  - Protects the user from errors (frequent errors, unhelpful error messages, and software crashes can frustrate users/customers, and this is not good)
  - Protects the database from errors (data can be lost or overwritten)
  - Errors can be costly, in time and resources (processes may slow as operations are repeated or errors are investigated)

Handling Exceptions with PL/SQL

- A block always terminates when PL/SQL raises an exception, but you can specify an exception handler to perform final actions before the block ends

```
DECLARE
  v_country_name countries.country_name%TYPE := 'Korea, South';
  v_elevation countries.highest_elevation%TYPE;
BEGIN
  SELECT highest_elevation INTO v_elevation
 FROM countries WHERE country_name = v_country_name;
EXCEPTION
  WHEN NO_DATA_FOUND THEN
  DBMS_OUTPUT.PUT_LINE ('Country name, ' || v_country_name || ',
  cannot be found. Re-enter the country name using the correct
  spelling.');
END;
```

Handling Exceptions with PL/SQL

- The exception section begins with the keyword EXCEPTION

```
DECLARE
  v_country_name countries.country_name%TYPE := 'Korea,
South';
  v_elevation countries.highest_elevation%TYPE;
BEGIN
  SELECT highest_elevation INTO v_elevation
 FROM countries WHERE country_name = v_country_name;
EXCEPTION
  WHEN NO_DATA_FOUND THEN
  DBMS_OUTPUT.PUT_LINE ('Country name, ' || v_country_name ||
  ', cannot be found. Re-enter the country name using the
  correct spelling.');
END;
```

ORACLE
Academy

Handling Exceptions with PL/SQL

- When an exception is handled, the PL/SQL program does not terminate abruptly
- When an exception is raised, control immediately shifts to the exception section and the appropriate handler in the exception section is executed
- The PL/SQL block terminates with normal, successful completion

```
Country name, Korea, South, cannot be found. Re-enter the
country name using the correct spelling.

Statement processed.
```

ORACLE Academy

Only one exception can occur at a time and only one handler (although it may include multiple statements) may be executed.

Handling Exceptions with PL/SQL

- The code at point A does not execute because the SELECT statement failed

```
DECLARE
  v_country_name countries.country_name%TYPE := 'Korea, South';
  v_elevation countries.highest_elevation%TYPE;
BEGIN
  SELECT highest_elevation INTO v_elevation
 FROM countries WHERE country_name = v_country_name;
  DBMS_OUTPUT.PUT_LINE(v_elevation); -- Point A
EXCEPTION
  WHEN NO_DATA_FOUND THEN
  DBMS_OUTPUT.PUT_LINE ('Country name, ' || v_country_name || ',
  cannot be found. Re-enter the country name using the correct
  spelling.');
END;
```

ORACLE
Academy

Handling Exceptions with PL/SQL

- When an exception is raised, the rest of the executable section of the block is NOT executed; instead, the EXCEPTION section is searched for a suitable handler

Handling Exceptions with PL/SQL

- The following is another example
- The select statement in the block is retrieving the last_name of Stock Clerks

```
DECLARE
  v_lname VARCHAR2(15);
BEGIN
  SELECT last_name INTO v_lname
 FROM employees WHERE job_id = 'ST_CLERK';
  DBMS_OUTPUT.PUT_LINE('The last name of the ST_CLERK is :
'||v_lname);
END;
```

```
ORA-01422:  exact fetch returns more than requested number of rows
```

- However, an exception is raised because more than one ST_CLERK exists in the data

ORACLE
Academy

There is no exception handler in this block, therefore the block terminates unsuccessfully, returning an "unhandled exception" status code to the calling environment (Application Express), which then reports the exception as shown.

Handling Exceptions with PL/SQL

- The following code includes a handler for the predefined Oracle server error called TOO_MANY_ROWS
- You will learn more about predefined server errors in the next lesson

ORACLE Academy

This code will successfully handle the exception inside the block, so PL/SQL returns a "success" status code to the calling environment, which therefore will report Statement Processed (below the display of the PUT_LINE).

Handling Exceptions with PL/SQL

```
DECLARE
  v_lname employees.last_name%TYPE;
BEGIN
  SELECT last_name INTO v_lname
 FROM employees WHERE job_id = 'ST_CLERK';
  DBMS_OUTPUT.PUT_LINE('The last name of the ST_CLERK is: '
|| v_lname);
EXCEPTION
  WHEN TOO_MANY_ROWS THEN
 DBMS_OUTPUT.PUT_LINE ('Your select statement retrieved
                  multiple rows. Consider using a
                  cursor.');
END;
```

ORACLE
Academy

Trapping Exceptions

- You can handle or "trap" any error by including a corresponding handler within the exception-handling section of the PL/SQL block
- Syntax:

```
EXCEPTION
  WHEN exception1 [OR exception2 . . .] THEN
 statement1;
 statement2;
 . . .
  [WHEN exception3 [OR exception4 . . .] THEN
 statement1;
 statement2;
 . . .]
  [WHEN OTHERS THEN
 statement1;
 statement2;
 . . .]
```

Trapping Exceptions

- Each handler consists of a WHEN clause, which specifies an exception name (exception1, exception 2, etc.), followed by THEN and one or more statements to be executed when that exception is raised (statement1, statement 2, etc.)
- You can include any number of handlers within an EXCEPTION section to handle different exceptions

Trapping Exceptions

```
EXCEPTION
  WHEN exception1 [OR exception2 . . .] THEN
 statement1;
 statement2;
 . . .
  [WHEN OTHERS THEN
 statement1;
 statement2;
 . . .]
```

Trapping

- In the syntax, OTHERS is an optional exception-handling clause that traps any exceptions that have not been explicitly handled

```
EXCEPTION
  WHEN exception1 [OR exception2 . . .] THEN
 statement1;
 statement2;
 . . .
  [WHEN OTHERS THEN
 statement1;
 statement2;
 . . .]
```

The OTHERS Exception Handler

- The exception-handling section traps only those exceptions that are specified; any other exceptions are not trapped unless you use the OTHERS exception handler
- The OTHERS handler traps all the exceptions that are not already trapped
- If used, OTHERS must be the last exception handler that is defined

ORACLE Academy

The OTHERS Exception Handler

- Consider the following example:

```
BEGIN
  ...
  ...
EXCEPTION
  WHEN NO_DATA_FOUND THEN
  statement1;
  statement2;
  ...
  WHEN TOO_MANY_ROWS THEN
  statement3;
  statement4;
  ...
  WHEN OTHERS THEN
  statement5;
  statement6;
  ...
END;
```

ORACLE
Academy

PLSQL 7-1
Handling Exceptions

26

If the exception NO_DATA_FOUND is raised by the program, then the statements in the corresponding handler are executed.

If the exception TOO_MANY_ROWS is raised, then the statements in the corresponding handler are executed.

However, if some other exception is raised, then the statements in the OTHERS exception handler are executed.

Guidelines for Trapping Exceptions

- Follow these guidelines when trapping exceptions:
  - Always add exception handlers whenever there is a possibility of an error occurring
  - Errors are especially likely during calculations, string manipulation, and SQL database operations
  - Handle named exceptions whenever possible, instead of using OTHERS in exception handlers
  - Learn the names and causes of the predefined exceptions
  - Test your code with different combinations of bad data to see what potential errors arise

Guidelines for Trapping Exceptions

- Write out debugging information in your exception handlers
- Carefully consider whether each exception handler should commit the transaction, roll it back, or let it continue
- No matter how severe the error is, you want to leave the database in a consistent state and avoid storing any bad data

ORACLE
Academy

Terminology

- Key terms used in this lesson included:
  - Exception
  - Exception handler

- Exception – Occurs when an error is discovered during the execution of a program that disrupts the normal operation of the program.
- Exception Handler – Code that defines the recovery actions to be performed when execution-time errors occur.

Summary

- In this lesson, you should have learned how to:
  - Describe several advantages of including exception handling code in PL/SQL
  - Describe the purpose of an EXCEPTION section in a PL/SQL block
  - Create PL/SQL code to include an EXCEPTION section
  - List several guidelines for exception handling

ORACLE
Academy